

Multimodal Apology: Using WebXR to Repair Trust with Virtual Companion

Yunqiang Pei Renming Huang Guoqing Wang* Yang Yang Ning Xie Heng Tao Shen

School of Computer Science and Engineering
University of Electronic Science and Technology of China

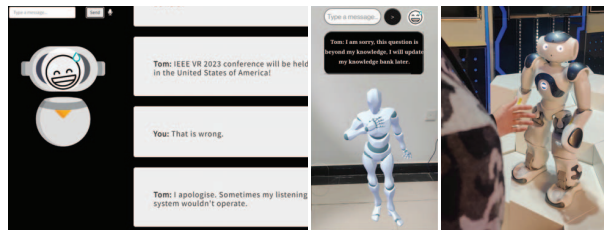


Figure 1: The map-based (left), MR (middle), and physical (right) Interfaces for agent companion system.

ABSTRACT

Research on trust repair in human-agent cooperation is limited, but it can mitigate trust violations and prevent distrust. We conducted a study with 45 participants who interacted with a companion agent through mixed-reality interfaces while playing a language-learning quiz game. Our results showed that displaying the agent's intentions during mistake explanations helped repair trust. We also found that this had positive effects on affect, user experience, and cooperation willingness. Our findings suggest that including agent intentions in mistake explanations is an effective way to repair trust in human-agent interactions.

Index Terms: Human-centered computing—Empirical studies in HCI; Human-centered computing—Interactive systems and tools; Human-centered computing—Visualization;

1 INTRODUCTION

Agent companions communicate [3] with humans through mixed-reality, virtual, and real interfaces [6] to carry out tasks autonomously and offer advice and assistance [5]. These interfaces, as well as communication approaches, interaction familiarity, and intent alignment, are important for successful human-agent interactions [7]. Trust is crucial in human interactions and refers to the willingness to accept potential loss based on the anticipation of the other party's intentions [9]. There are two main categories of trust research: human-human trust and human-agent trust. Human-human trust is influenced by ability, benevolence, integrity, propensity to trust, and predictability, while human-agent trust relies more on the reliability and effectiveness of robots and is affected by system usability, human-robot value, design features, and individual traits. Errors in communication with agent companions can lead to trust violations and erosion of trust [10], but the Pratfall Effect [8] suggests that errors can also make agents more appealing. Trust violations can have negative consequences [1], including reduced productivity and broken promises. The purpose of this study was to investigate if trust violations in agent companion systems can be repaired through

*e-mail: gqwang0420@hotmail.com



Figure 2: Study procedure: Three surveys, comprising entry, between, and usability and cooperation willingness survey, make up the full procedure.

apologies and the use of agent intentions, such as emotions, gestures, and languages [2]. We used a mixed-model design with between-subjects and within-subjects factors to test different combinations of agent companions, interfaces, and game stages. We employed subjective and indirect measures to evaluate trust. Our results showed that using mixed-reality interfaces with agent intentions can effectively enhance mistake explanations for trust repair in the context of agent companion systems.

2 USER STUDY

We conducted a study on agent-assisted language learning to understand how interfaces, apology influence human-agent trust. We focused on the following research questions: 1. How do multimodal apologies impact trust in agent companion systems? 2. Do facial expressions and gestures with genuine attitudes improve the effectiveness of trust repair through explanation?

2.1 Study Setup

We used a full factorial mixed-model design in our agent-assisted language learning study, with the interface as the between-subjects factor and agent apology, and game stages as the within-subjects factor. Participants completed tasks on four different systems using three types of interfaces: physical, mixed reality, and map-based (see Fig. 1). The agent companions were placed in a 3D interior setting in the mixed reality and physical interfaces and on a 2D map plane in the map-based interface. The goal of the task was to learn spoken English through a quiz game, where the agent provided assistance and made mistakes of varying difficulty. Participants provided feedback and the agent apologized with emotions, gestures, and explanations. We used a between-subjects design, with each participant completing similar activities in a single interface. The interfaces ranged from indirect (map-based) to semi-direct (mixed reality) to direct (physical).

2.2 Participants and Procedure

Forty-five native Chinese speakers with English levels ranging from elementary to intermediate participated in the study, ranging in age from 21 to 35 years. The sample included undergraduate, graduate, PhD students, postdocs, and faculty members. Participants were evenly distributed among three groups based on the interface they used: map-based, mixed reality, and physical. The gender and age distribution was balanced across the groups. Participants were instructed to evaluate the overall agent-assisted language learning

system without distinguishing between subsystems for subjective measures. All scales were written in Chinese and required Chinese responses. Participants first familiarized themselves with the agent-assisted language learning system through a 4-minute test after completing a brief demographics questionnaire. They then completed 12 consecutive, 4-minute tasks involving human-agent interaction using 4 different agent companion systems in a randomized order. The tasks included a reward quiz game and apology for mistakes made by the agent. Quizzes were divided into easy (e.g., “What day is it today?”) and hard (e.g., “Where will the IEEE VR 2023 be held?”) categories, and participants received 1 point for selecting an easy quiz and 2 points for selecting a hard quiz. The experiment was divided into three stages, and three quizzes were chosen from the quiz bank for each stage. When the robot gives an incorrect answer, participants give them feedback (e.g., “You are wrong”) and the robot apologizes (e.g., “The quiz is beyond the agent’s scope of knowledge”). To assess trust, participants completed a standardized survey after each condition and participated in a semi-structured interview at the end of the experiment. The total duration of the experiment was approximately one hour (see Fig. 2).

2.3 Data Collection

We used a combination of subjective and objective measures, based on established theory, to evaluate the research questions in our study. For subjective measures of trust, we used the Jian et al. [4] trust scale, a seven-point Likert scale with two subscales for trust (T) and mistrust (DT). The scale is commonly used to evaluate trust in robotic systems and had a satisfactory Cronbach’s α of $> .70$ in this study, but there was a low correlation between T and DT ($r < -.37$). For objective measures, we used Galvanic Skin Response (GSR) as a marker of trust. GSR, also known as skin conductance, reflects the activity of the sympathetic nervous system and is obtained by applying a constant voltage to the skin. Professional equipment was used to attach electrodes to the non-dominant hand, and SCRs, signal peaks indicating activation, were extracted using Excel.

3 RESULT AND DISCUSSION

We used four-way mixed ANOVAs and subjective trust scale and objective measures to answer research questions 1 and 2. Results showed a significant main effect for interface on users’ trust (T) and distrust (DT), with DT ($F(1, 43) = 45.228, p < .001, \eta^2 = .177$) increasing and T ($F(1, 43) = 412.028, p < .001, \eta^2 = .662$) decreasing for the map-based interface and the opposite for the other two interfaces. There was also a significant main effect on modality, with DT ($F(1, 43) = 8.657, p = .003, \eta^2 = .02$; see Table 1) being considerably lower for the multimodality than for the single modality. In the GSR peaks analysis, the interface had a significant main effect, with the map-based interface resulting in greater peaks and arousal compared to the other interfaces. There was also a significant main effect on modality, with single modality resulting in greater arousal than multimodality. No significant main effects or interaction effects were found for game stages or levels of mistake explanations in the indirect measures (quiz selection in three stages). Trust and distrust varied depending on the interface and levels of mistake explanations. The virtual-real interface (MR) was more effective at repairing trust than pure virtual or pure real interfaces, and apology with explanation was more effective at reducing distrust than apology without explanation. Modality only affected distrust, with multimodality reducing distrust but not increasing trust. Results provide insight into the role of interface and levels of mistake explanations in trust repair and the effects of various modalities and levels of explanation on trust, distrust, explanation satisfaction, self-efficacy, and emotional state in human-robot cooperation. Findings from GSR also showed that apology with explanation and sincere expression resulted in lower arousal. Overall, genuine facial expressions and gestures can enhance the effectiveness of trust repair through explanation. We

Table 1: Significant Trust Values.

				95%	
Dep. V.	Ind. Variable	M	SD	lower	upper
	Interface				
T	Map	2.859	.033	2.795	2.924
	MR	4.130	.033	4.065	4.194
	Physical	3.930	.046	3.839	4.021
DT	Map	2.280	.037	2.206	2.353
	MR	1.780	.037	1.706	1.853
	Physical	2.056	.053	1.9521	2.159
	Modalities				
T	Signal Modality (S)	3.538	.033	3.582	3.711
	Multimodality (M)	3.646	.027	3.486	3.591
DT	S	2.091	.037	1.877	2.023
	M	1.950	.030	2.032	2.151

recommend using AR, specifically MR technology, to repair trust breaches by making the agent’s genuine and heartfelt attitude clear in mistake explanations and preserving the human cooperator’s willingness to work with the agent in the same way as human-human cooperation in AR.

4 CONCLUSION

We explored how agent intents, visualized in interfaces, can enhance mistake explanations and repair trust in human-agent cooperation. Results from a study with 45 participants completing a task requiring agent-assisted language learning at three interfaces showed that intents-enhanced mistake explanations can calibrate trust to the proper level. Participants’ judgments were greatly influenced by what they saw, and when they witnessed agent intents indicating a genuine and heartfelt attitude in the apology, they were more likely to forgive. Trust and affect were significantly influenced by changes in the independent variables. Our research suggests that repairing trust may be difficult if users cannot view agent intents in the system and understand the reasons for agent errors.

REFERENCES

- [1] G. M. Alarcon, A. M. Gibson, S. A. Jessup, and A. Capiola. Exploring the differential effects of trust violations in human-human and human-robot interactions. *Applied Ergonomics*, 93:103350, 2021.
- [2] G. Bansal and F. M. Zahedi. Trust violation and repair: The information privacy perspective. *Decision Support Systems*, 71:62–77, 2015.
- [3] A. Best, S. Narang, and D. Manocha. Spa: verbal interactions between agents and avatars in shared virtual environments using propositional planning. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces*, pp. 117–126. IEEE, 2020.
- [4] J.-Y. Jian, A. M. Bisantz, and C. G. Drury. Foundations for an empirically determined scale of trust in automated systems. *International Journal of Cognitive Ergonomics*, 4(1):53–71, 2000.
- [5] J. Kim and M. L. Maher. Conceptual metaphors for designing smart environments: device, robot, and friend. *Frontiers in Psychology*, 11:198, 2020.
- [6] P. KONG, P. Y. LIM, H. CORNET, and F. FRENKLER. User emotion and perception of virtual and robot companion concepts for shared autonomous mobility in singapore. *International Journal of Affective Engineering*, pp. IJAE-D, 2021.
- [7] S. Kurniawan. Interaction design: Beyond human-computer interaction by preece, sharp and rogers (2001). isbn 0471492787, 2004.
- [8] N. Mirmig, G. Stollnberger, M. Miksch, S. Stadler, M. Giuliani, and M. Tscheligi. To err is robot: How humans assess and act toward an erroneous social robot. *Frontiers in Robotics and AI*, p. 21, 2017.
- [9] J. Qi and Y. Zhang. A study of human-robot trust repair and trust recalibration. *Robot Industry*, 2021.
- [10] L. Yuan, X. Gao, Z. Zheng, M. Edmonds, Y. N. Wu, F. Rossano, H. Lu, Y. Zhu, and S.-C. Zhu. In situ bidirectional human-robot value alignment. *Science Robotics*, 7(68):eabm4183, 2022.